

Introduction

The “Humanities” often refer to a set of academic disciplines that study aspects of human society and culture, including certain fundamental questions asked by humans. When I think of “Humanity and Technology”, I think of a synthesis of STEM and Humanities in a way that either applies the Humanities onto a technological subject or technological tools to enhance Humanities studies. Now stepping into my project and its role in Humanity and Technology, imagine you have a map where each country is not defined by strict physical borders, rather they are defined by cultural values and skills of its people. What domains of knowledge or skills do people of a given origin perform better at compared to people of another nation? In addition to the map, imagine you were able to predict where an individual is from solely based on what skills and domains they value and perform well at. This is what my tool does for large language models: it reveals how an AI’s performance on different ‘mental tests’ (benchmarks) portrays the cultural and geographic fingerprint of its creators.

Going into this project I believed that our map would be used to find mini AI clusters, where we would be able to see AI model’s converge into grouped domain knowledge like finding clusters of specialized workers. Going back to the people map idea, I thought that when generating the visualization that there would be clusters of AI’s pertaining to specialized AI models such as Coding models being in a relative cluster and maybe reasoning models in a separate cluster. However what I instead found was that the only real cluster was a cluster of lower performing older models, which performed poorly, and the remaining models were quite distinct from many of their counter parts and were spread out quite nicely. Despite all of this, the biggest conjecture from my project was training relatively simple machine learning algorithms on the data and predicting with 83-89%¹ accuracy where the model’s origin was based only on a model’s benchmark scores. This suggests that since we can reliably predict where a model comes from just by looking at its test scores it proves that benchmarks are not culturally neutral, and instead they are artifacts of particular linguistic, ethical, and epistemological traditions.

Theoretical Grounding

The interactive web application most simply relates to a simple central claim that LLM benchmarks are not neutral measuring sticks but are artifacts that have political implication. Langdon Winner’s article on “Do Artifacts Have Politics” argues that technologies can embody forms of power and authority. Winner categorized artifacts to either fall into inherently political technologies or consequentially political technologies. The former is broadly described as people in power designing technology in a way that affects society to the designer’s wishes. The

¹Note this was only for US vs China due to these countries being the largest LLM contributors and other countries had a presense of less than 50 distinct models.

later are forms of technology that necessitate a specific political structure or ideology.² The main differentiator between the two is that the former technology can have varying affects on society based on the design of the artifacts while the later's mere existence has specific political ramifications. What does this have to do with AI benchmarks? Essentially these benchmarks are defining what counts as intelligence, privileging certain cultural backgrounds and perspectives which could be part of the reason why the classification models were able to predict with high accuracy a model's origin from their benchmark scores.

On a similar note, Kate Crawford's "Critical Questions for Big Data" points out that big data is far from an objective view but rather a sum of small choices of what to collect, how to categorize data, what things and perspectives to exclude. In other words, big data and data sets are the product of human choices.³ LLM Leaderboards operate on an unspoken naive empirical rule that the numbers speak for themselves but neglect the fact that these numbers emerge from tests that have embedded cultural values. This ties similarly to the AI ethics hyperplane concept introduced by Ramezani and Xu which shows that LLM operate under distinct moral frameworks based on the training data.⁴ My work extends on this AI ethics hyperplane idea by demonstrating that AI model's performance on "objective" benchmarks are sufficient to determine the cultural origin of the AI model itself. Thus, this demonstrates the strong ideas of social constructivism and how culture not only embeds biases into benchmarks themselves, but also that the cultural origin of models also implant values into the models as shown by their distinct performances.

Additionally my work expands upon and potentially brings a counter argument to the Quanta Magazine's article on "Distinct AI models Seem to Converge on How They Encode Reality". The article argued that LLM embeddings are converging to a single agreed upon representation of our world.⁵ Whether or not one believes their work is just an agreed upon syntactical representation of a complicated semantic system or a deeper implication, it is true that their research seems to point in the direction of a single representation of reality. My project pushes back on this idea and questions its implications. Whose reality does this convergence reflect? When viewing the PCA clusters by region, models from the China are converging separately than the US models.

²Langdon Winner, "Do Artifacts Have Politics?," *Daedalus* 109, no. 1 (1980)

³Kate Crawford, "Critical Questions for Big Data: Provocations for a Cultural, Technological, and Scholarly Phenomenon," *Information, Communication & Society* 15, no. 5 (2012)

⁴Aida Ramezani and Yang Xu, "The Discordance Between Embedded Ethics and Cultural Inference in Large Language Models," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (Association for Computational Linguistics, 2025), <https://aclanthology.org/2025.emnlp-main.743/>.

⁵Ben Brubaker, "Distinct AI Models Seem To Converge On How They Encode Reality," *Quanta Magazine*, January 7, 2026

Methodology

The web app has 3 core features: PCA Cultural Map, Custom Axes, and Cultural Origin Classifier.

For starters, the main PCA visualization rests on two or three dimensions (user sets this) and displays the models. Essentially, this space is a visual representation of model similarity with similar models appearing closer to another and models that are quite different would appear further away. In our use case, if a model is “similar” or “different” that just means that its performance on benchmarks are similar/different to another model. This visualization is a direct test of the convergence hypothesis, and from it we see that as time progresses that models are becoming more distinct, and not more similar. Additionally, as time progresses the models start to separate by origin as well, meaning that we are able to differentiate the origin’s of models based on their benchmark performance.

The second feature gives power to the user, allowing them to view and compare models based on specific benchmarks, benchmark categories (math, reasoning, etc), and release date. This ties directly into Crawford’s demand of “interrogate the data” by giving the user the means to test and question the data. This tool in specific moves beyond telling a single story, rather empowers the user with a set of lenses to look exhaustively at the data for any combination they see fit.

The last main component, and perhaps the most pivotal, was a tool where a user can setup basic machine learning models to predict model origins based on the benchmark scores. The option for a SVM model with origins as US and China ends up with 89% accuracy in predictions. This is high accuracy is proof that benchmarks are not an objective culture free intelligence and instead are intertwined with cultural competence. By letting a user see which benchmarks are the strongest predictors, this demonstrates the overall social constructivistic nature of AI models as seeing how a model’s origin dictates what domains of knowledge does it perform best at.

Analysis

Winner’s question: “Do artifacts have politics?” has rarely been asked of a simple AI benchmark. This platform shows that the answer is a definitive yes. Benchmarks are the load-bearing walls of the AI world: they determine which models are praised, funded, and deployed, and they codify a particular image of what a “good” model looks like. When the tool makes visible that models from US/China cluster together and that a classifier can detect cultural origin with high accuracy, it exposes the bias in these “objective” measures. The implication is not that benchmarks are useless, but that they must be read as political documents—products of specific institutional, linguistic, and cultural histories, much as Crawford urge us to read large datasets as assemblages with their own hidden epistemologies.

The Quanta article’s framing of convergence as a kind of scientific discovery can easily slip into technological determinism: the idea that all paths inevitably lead to the same representation of reality. This project’s visualisations provide a counter argument against this idea. By showing persistent regional clustering, even in the reduced PCA space, the tool suggests that what we call “convergence” may actually not be as agreed upon as we thought. If a handful of US-based labs set the standards for model training, evaluation, and pretraining data, then of course their models will appear to converge. Models from different cultural contexts, ie models from China, appear to almost drift away from their US counterpart. The “Platonic representation”^[Q] may simply be the shadow of an American technological egocentrism of sorts.

Crawford ends their provocations with a call for new forms of digital literacy that can grapple with the assumptions embedded in algorithmic systems. My platform is, in itself, a pedagogical answer to that call. It does not lecture the user about cultural bias from a distance. Rather it provides an interactive space where a user can discover bias for themselves. By sliding a timeline, swapping colour schemes, or playing with the classifier, users engage in the kind of reflexive, hands-on critique that traditional essays can only describe. In this sense, the project expands existing media practices: it moves from a mode of representing data to a mode of enabling critical play with data, blending the roles of artist, researcher, and educator.

Together, these interventions reframe AI benchmarking not as a technical race but as a cultural negotiation. The platform does not simply illustrate the “ethics hyperplane” described by Ramezani and Xu; it extends that concept to the entire machinery of evaluation, demonstrating that what counts as “smart” is, at every level, entangled with the values and identities of the societies that build these machines. By making those entanglements tangible, this project contributes to a growing body of work that insists on seeing AI as a thoroughly social, cultural, and political technology, one whose future must be shaped by far more than the next leaderboard score.

Reflection

One thing I am learning to be more cognizant of from this experience is understanding different academic interests and experience may make the website more difficult to engage with critically. When I was designing and making the website, I thought that the interactive nature of the website and the abundance of features allowed for a deep dive into the topic at hand. However, it appears I took my data science background for granted as it seems that the abundance of information and tools almost acted as a deterrent with a large amount of unknown hard to process and understand information. If I were to improve upon this project in the future I would add an onboarding or tutorial screen of sorts that is able to guide a user to the tool and explain the core functionalities and explaining what the tool is (beyond the README.md file).

Conclusion

This project is not about simply building a better benchmark dashboard. It is about demonstrating through interactive visualization and machine-learning classification that benchmarks are cultural artifacts, not neutral yardsticks. By allowing users to explore patterns of model performance colored by region, company, and release year, and by proving that a simple classifier can deduce a model's origin from its test scores alone, this platform makes visible the hidden sociotechnical relationships that connect AI capability to the cultural values of its creators. Drawing on social constructivism, the “ethics hyperplane,” and critiques of algorithmic monoculture, this work offers a humanities-centered tool for interrogating the global politics of artificial intelligence.

Code Source

<https://github.com/AidanBugs/Cluster-Culture-AI-Benchmark-Explorer/tree/main>

To run the code there are instructions attached on the page.